PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME					
CT GROUP	1	6	S		
INDEX NUMBER					

BIOLOGY 9744/03

Paper 3 Long Structured and Free-response Questions **18 Sep 2017**

2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate. All workings and appropriate units must be shown.

The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of 2 sections:

Section A:
Answer **ALL** questions.
Answers are to be written in the spaces provided.

Section B:
Answer **ONE** question.
Write your answers on the separate writing papers provided.

Please hand in section A and section B separately.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	**/ 28**
2	**/ 10**
3	**/ 12**
Section B	
4 or 5	**/ 25**
Total	**/ 75**

This document consists of **15** printed pages including the cover page and **1** blank page.

Section A

Answer **all** questions in this section.

1 Mosquito vectors *Aedes aegypti* and *Aedes aldopictus* are main vectors of dengue virus (DENV) and chikungunya virus worldwide.

With about 50-100 million reported cases annually, including 500 000 severe cases of dengue haemorrhagic fever (DHF) or dengue shock syndrome (DSS), DENV is the most prevalent mosquito-borne human virus worldwide.

(a) Outline the life cycle of *Aedes aegypti*. [2]

a. Female mosquitoes lay (100-200) eggs inside water bodies (over several sites)
b. When the eggs are submerged in water, the eggs hatch and larvae emerges.
c. The larvae are mainly found at the water's surface, feeding on organic particulate matter such as algae.
d. (After as little as 5 days) The larvae develop into pupae.
e. After 2 days, adults emerge head-first by ingesting air to expand the abdomen and thus splitting open the pupal case.

Fig. 1.1 shows a graph on development time of immature mosquitoes to the adult reproductive stage. The data is based on studies in a laboratory with mosquitoes taken from a tropical forest.

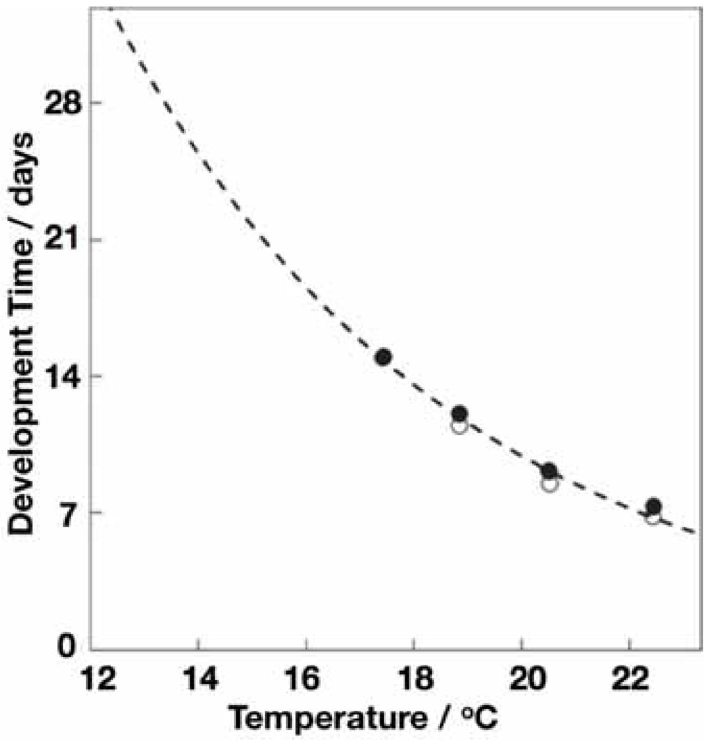

Fig. 1.1

(b) With reference to Fig. 1.1,

(i) explain how temperature changes impacts insects' metabolism. [2]

a. (Quote data) As temperature increases, the developmental time from larvae to adult decreases
b. It means that there is faster emergence of adult

c. Mosquitoes are poikilothermic;
d. Metabolic processes are enzyme-mediated processes;
e. Hence, increase in temperature (towards optimum temperature) corresponding increases metabolic process that leads to growth and development;

(ii) explain the consequence of the trend on the spread of dengue virus. [2]
a. Since developmental time is shortened, greater survival rate of larvae as less susceptible to predators, diseases and parasitism
b. -> greater population of mosquitoes -> more vectors to carry the virus
c. Global warmer also results in warmer and shorter winters, allowing more mosquitoes to survive during and through winter as they can be active for a longer period of time
d. Mosquitoes can also migrate polewards as regions towards the poles are also becoming warmer -> spread of dengue virus latitudinally

The immune system is the body's defense against infectious organisms and other invaders. Through a series of steps called the immune response, the immune system attacks organisms and substances that invade body systems and cause disease.

(c) **(i)** Describe how the innate immune system normally responds to a microbial infection in the skin tissue, such as DENV. [3]
a. (tissue-resident) DCs / Langerhans cells and macrophages recognise the PAMPS on the microbes by their PRRs;
b. resulting in (receptor-mediated) phagocytosis of the bound microbe;
c. phagocyte becomes activated □ secrete cytokines;
d. which aids in the recruitment of circulating phagocytes (monocytes / neutrophils) from the blood circulation into the infected tissue;
e. resulting in inflammation;
f. recruited phagocytes assist in the elimination of the microbes at the site of infection;

(ii) Explain why the normal innate immune responses prove to be ineffective when the body is infected with DENV. [2]
a. Langerhans cells and macrophages are the host / target cells for DENV;
b. (idea of) phagocytosis allows for entry of virus into the host cell;
c. virus able to escape lysosomal degradation and replicate within the host cell/AW;
d. resulting in increase in DENV rather than eliminating DENV/AW;

(d) In response to the DENV infection, the body's immune response reacts by producing antibodies that target the DENV virus.

(i) Explain how the structure of the antibody allows for the successful recognition and binding of DENV virus in blood. [2]
a. contains two antigen-binding sites;
b. each is composed of one V_H and one V_L domain that is precisely folded to give a 3D conformation;
c. that is complementary to the antigens / proteins on the DENV;
d. allowing for the formation of weak / reversible bonds to be formed between the DENV and the antibody;

(ii) Describe two ways in which the production of antibodies help in removing DENV from the body in the primary antibody response. [2]

Choose any two pairs of answers:

a. neutralisation;
b. where the binding of the antibodies to the antigen prevents the interaction of the DENV to the host cell receptors;
c. opsonisation;
d. where the antibody-bound DENV is recognised by phagocytes, thus enhancing the clearance of the DENV from the blood;

e. agglutination;
f. where DENV bound by antibodies are concentrated → lesser infectious units → easier to clear DENV;

g. activation of complement proteins;
h. where the antibody-bound DENV recruits complement proteins that assemble on DENV surface → lysis;

(e) Despite the protection offered by the antibodies in the primary infection, the recurrent exposure to DENV, particularly of a different serotype, can result in the manifestation of severe dengue fever.

(i) State how the four different DENV serotypes differ. [1]

differ in their composition of their antigens ;;

(ii) Explain why the infection by a different serotype can result in severe dengue fever. [2]

a. due to antibody-dependent enhancement;
b. where the antibodies produced from the primary infection binds to the infective DENV virus during a subsequent infection / secondary infection;
c. the antibody-bound DENV is then recognised by (circulating) monocytes / macrophages by the Fc receptors;
d. resulting in entry of virus into the cell → replicate to large numbers → viremia → severe dengue fever;

In 2016, the Health Sciences Authority (HSA) has approved the world's first dengue vaccine Dengvaxia for use in Singapore. Dengavia is a live, attenuated tetravalent dengue vaccine, and has shown to be effective in causing protection against the four DENV serotypes.

(f) Discuss two advantages of vaccination in the eradication of diseases such as dengue. [2]

Advantage 1:
a. prevents the development of disease in uninfected individuals;
b. especially when in disease-prone areas / infected with the pathogen;

Advantage 2:
c. results in a reduction in disease transmission within the population;
d. due to herd immunity → little chances for disease outbreak;

Advantage 3:
e. vaccination is far cheaper as a solution to combat diseases;
f. compared to medical care which can incur a higher cost if the medical care requires an extended period of time;

Max 2m

Besides vaccination, vector control programmes are in widely adopted as a preventive measure. Unfortunately, these programmes are facing operational challenges with mosquitoes becoming resistant to commonly used insecticides in several areas through the world.

Spraying insecticide in regions with multiple stagnant water bodies is the main method of controlling *Aedes aegypti* in rural India. One of such insecticides was deltamethrin, which was introduced to rural areas in 2007.

A laboratory study was carried out using mosquitoes collected from two sites – A and B – in India. The percentage of mosquitoes killed by deltamethrin was estimated.

The results of the study are shown in Table 1.2.

Table 1.2

Site	Year	Percentage of mosquitoes killed by deltamethrin
A	2007	100
	2010	90
B	2007	100
	2010	100

(g) The researchers concluded that at Site **A**, the mosquitoes had evolved resistance to deltamethrin. Explain how the mosquitoes evolved resistance. [3]

a. **Mutation gave rise to genetic variation within the mosquitoes**
b. **-> some mosquitoes are resistant to deltamethrin while others are susceptible**
c. **Deltamethrin acts as a selection pressure**
d. **Mosquitoes which are resistant have a selective advantage**
e. **-> differential survival and reproductive rate**
f. **Allele coding for resistance is passed on to offspring**
g. **-> increase in frequency of resistance allele**
h. **And thus, natural selection has resulted in the evolution of resistance in mosquitoes.**
i.

India is one of the countries that has already been experiencing extreme weather events – extreme heat, droughts – due to climate change. Considering that agriculture play a vital role in India's economy, the impact of climate change on agricultural productivity has been a major concern.

Fig. 1.3 illustrates a prediction on global warming impacts on rice crop yield across India.

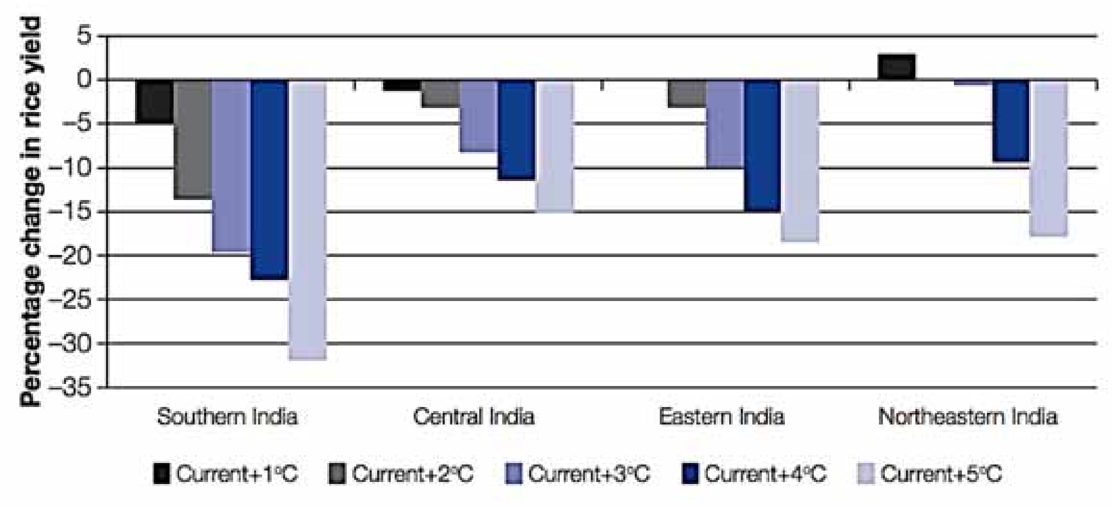

Fig. 1.3

In general, the trend is the similar for most plant crops.

(h) Explain the effects of increased temperature from climate change on plant crops. [2]

a. **As temperatures becomes warmer, crop yield decreases.**
b. **Extreme heat may impede the growth of plant crops.**
c. **The range and distribution of crop pests and weeds may increase, leading to extensive crop damages**
d. **Prolonged periods of heat may also place a stress on water supplies, which in turn dries the land and decreases crop yield**
e. **AVP**

It has been widely recognised that the effects of climate change have been brought about by excessive emission of greenhouse gases (GHG).

(i) A student made the following comment: 'If we stop deforestations, the concentration of GHG will decrease back to an acceptable level in the atmosphere'. Discuss the validity of this statement. [3]

a. **Yes, the concentration of GHG may decrease;;**
b. **Forests serve as natural carbon sinks**
c. **It also prevents over-exposure of the soil to accelerated rates of oxidation and decomposition which releases CO_2 and CH_4**

d. **No, besides deforestations, there are other main sources of emission of GHG;;**
e. **Burning of fossil fuels for energy required in powering transportations, machineries, homes**
f. **The increase in food consumption has also placed a greater demand on livestock production**
g. **Production and post-production of livestock is also a major source of GHG emission**
h. **It takes time for re-growth of forests that have already been removed, the current natural carbon sink may not be replenished fast enough**

For pt (d), award 1m if the idea of other sources of GHG is present in the rest of the answer

[Total: 28]

2 Fig. 2.1 shows the two distinct regions of human skin. The dermis is a thick region of living tissue below the epidermis, containing blood capillaries, nerve endings, sweat glands, hair follicles, and other structures.

The epidermis is composed of many different layers of different types of skin cells. These different layers of cells arose from the continual division and morphological changes of epidermal stem cells that are found in the basal layer of the epidermis.

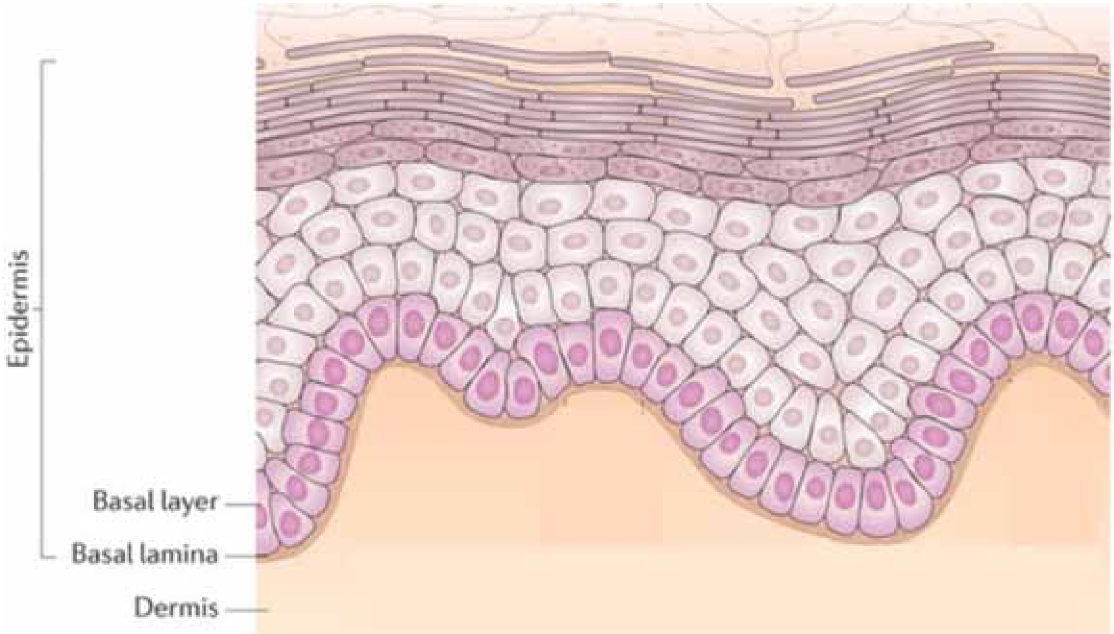

Fig. 2.1

(a) With reference to Fig. 2.1,

(i) describe the unique features of epidermal stem cells. [2]

a. **unspecialised cell that has no tissue-specific structures and functions;**
b. **capable of continuous cell division and self-renewal over a long period of time due to high telomerase levels;**
c. **producing genetically stable daughter cells / progeny;**
d. **which undergoes differentiation to give rise to skin cells which are pushed upwards;**

(ii) state the potency of these epidermal stem cells. [1]
multipotent;;

(iii) explain the importance of the epidermal stem cells in the skin. [2]
a. **maintains the architecture / thickness / structure of the skin tissue by replacing the cells of the epidermis ;;**
b. **particularly after injury / damage / abrasion of the upper epidermal layer ;;**

Until recently, burns have usually been treated with skin grafts, which involve taking skin sections from uninjured parts of the patient's body, and grafting them over the burn. These grafts can take several weeks or even months to heal, and during the recovery period, patients are prone to infections because of the damage to the skin, which is the body's first line of defence against microbes.

Scientists have now developed a new technique which involves harvesting the burn patient's skin stem cells, and stimulating them to divide using chemicals. These cells are then sprayed onto the burn. This method helps to regenerate the skin quickly, and dramatically reduce recovery times. Fig. 2.2 shows an illustration of this process.

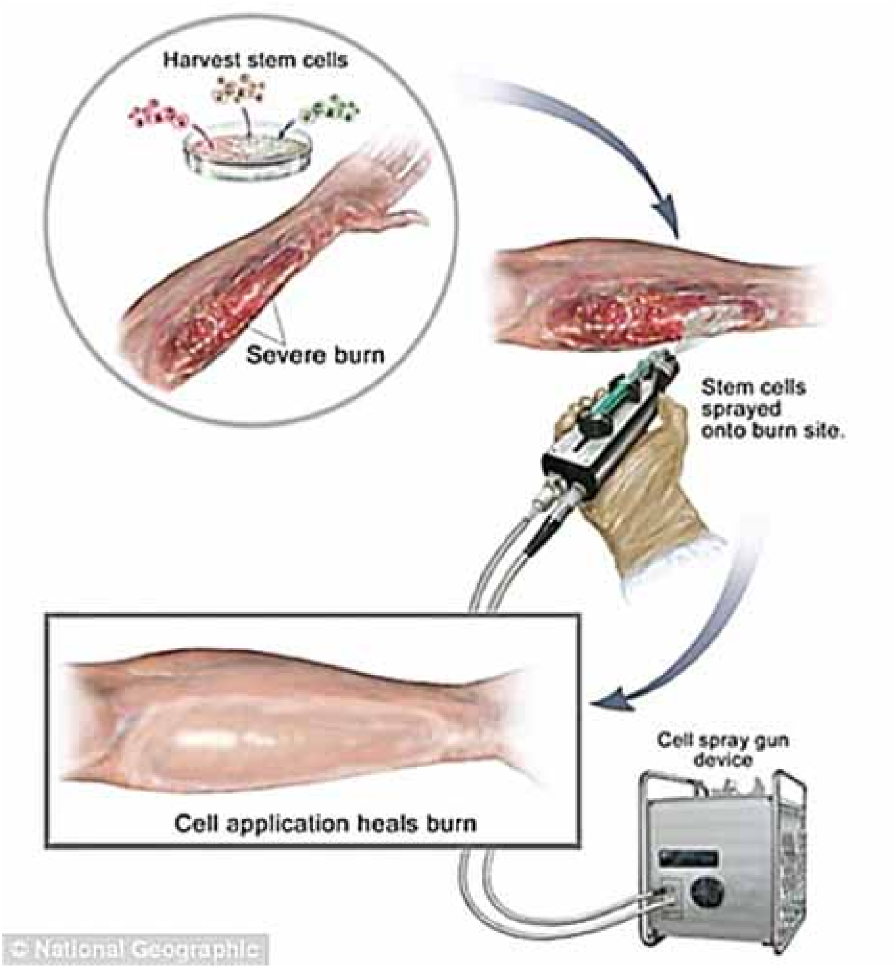

Fig. 2.2

Fig. 2.3 shows a photomicrograph of the skin stem cells undergoing repeated cell division in culture.

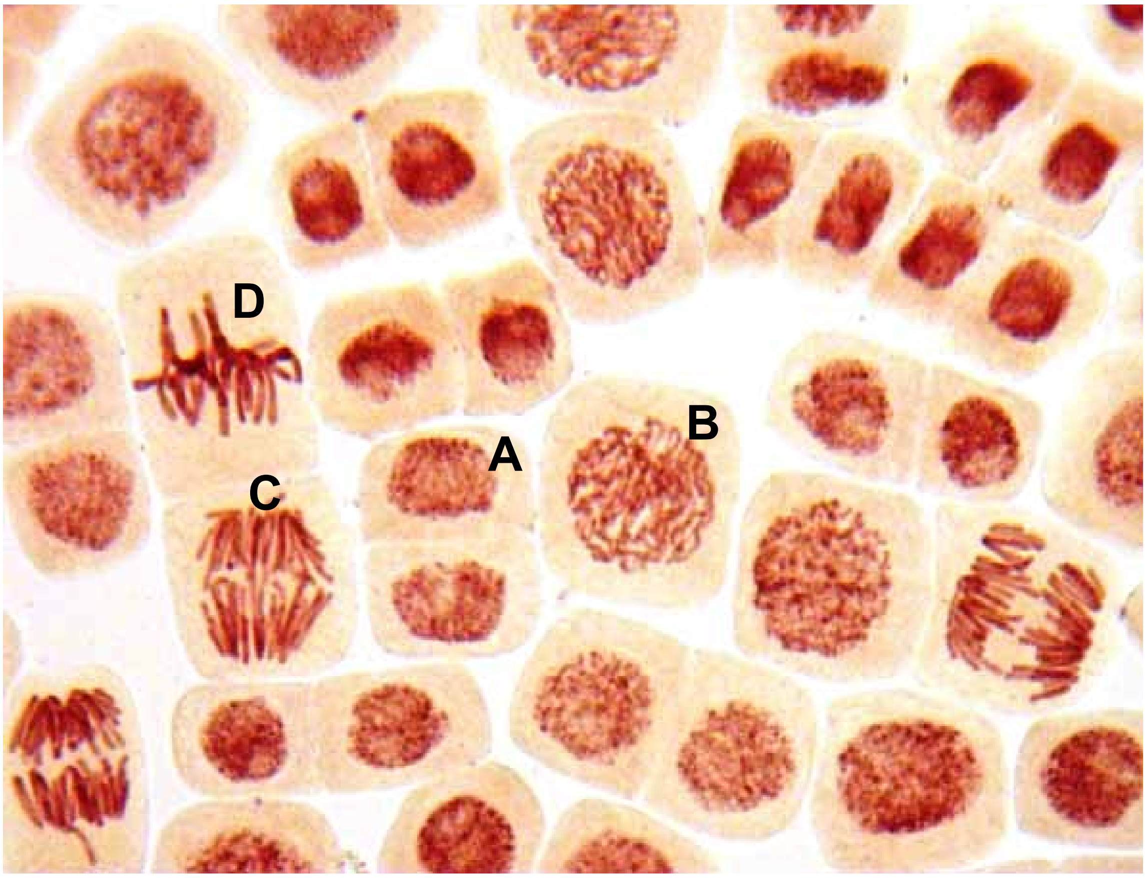

Fig. 2.3

(b) With reference to Fig. 2.3,

(i) arrange these stages in the correct sequence. [1]
B → D → C → A ;;

(ii) explain what is happening at stage **C**. [2]
a. <u>anaphase</u>;
b. MTs attached to kinetochore proteins on centromeres shorten;
c. while non-kinetochore MTs lengthen;
d. resulting in the separation of sister chromatids / centromeres holding sister chromatids separate → move to opposite poles, with the centromeres first;

(c) Suggest and explain why these stem cells need to be treated with chemicals to stimulate proliferation. [2]
a. epidermal stem cells are <u>adult stem cells</u> which are <u>generally quiescent</u>;
b. thus require presence of chemical signals to effect cell proliferation *in vivo*;
c. Removal of epidermal stem cells from patient's body → <u>absence of growth factors to stimulate cell division</u>;
d. Addition of chemicals stimulate cell division by binding to cell receptors and stimulating cell division pathways;

[Total: 10]

3 Fig. 3.1 shows a Siberian husky. The natural habitat of a Siberian husky is a cold, northern climate such as the Siberian Tundra or the wilds of Alaska. Siberian huskies were originally bred by the Chukchi people of Siberia to ultimately pull sleds across miles and miles of frozen ground. Basically, they were bred to be working dogs, as well as herd animals and perform as watchdogs.

Fig. 3.1

Fig. 3.2 represents the various birth weights of new-born puppies in a wild population of Siberian husky in Siberian Tundra. The line diagram on Fig. 3.2 represents mortality in relation to birth weight.

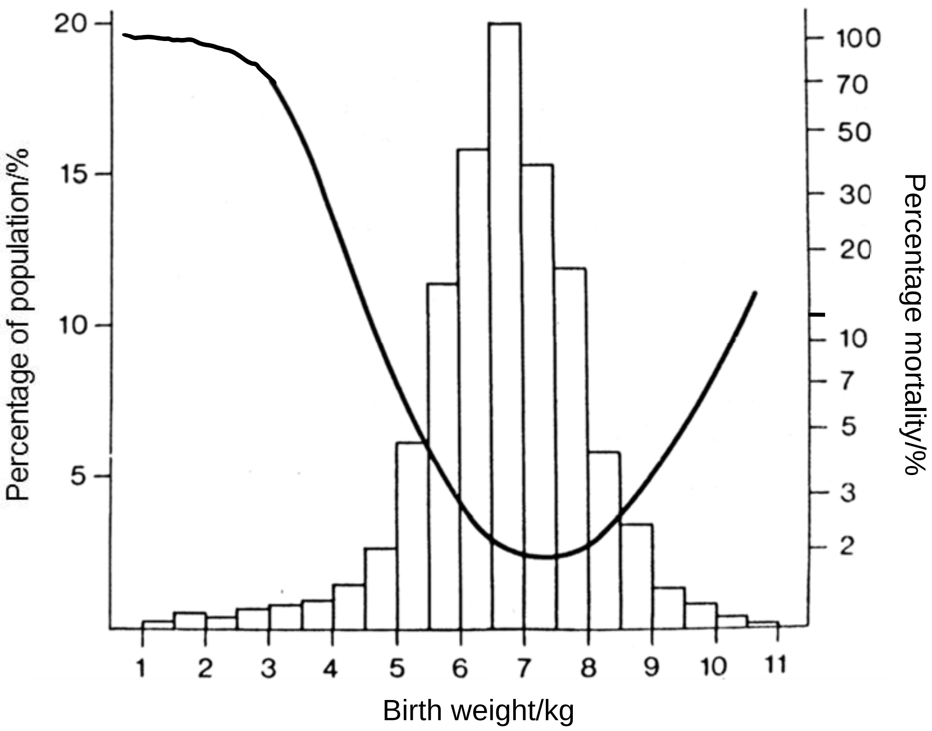

Fig. 3.2

(a) Using the information provided in Fig. 3.2, account for the type of selection acting on birth weight of new-born Siberian husky puppies. [3]

a. **stabilizing selection;;**
b. **Puppies with high and low birth weight are selected against / lower reproductive success, hence puppies with high and low birth weight have the lowest percentage population / highest mortality;**
c. **[Quote] low birth weight of 1kg -4.5kg → 0.5% - 2.5% of population / 98% to 10% mortality, high birth weight of 8.5kg – 11kg → 0.2% - 3% of population / 3% to 18% mortality [nos quoted are flexible]**
d. **Puppies with median birth weight of 5kg to 8.5kg are selected for / higher reproductive success. puppies with median birth weight has highest percentage population / lowest mortality.**
e. **[Quote] median birth weight of 5kg to 8.5kg → 6% increased to 20% and decreased to 6% population / more than 6% population / less than 5% . [nos quoted are flexible]**

(b) Birth weight of new-born Siberian husky puppies is an example of continuous variation. Explain why there is a variation of birth weights in the population of Siberian husky. [3]

a. **Birth weight are controlled by a large number of genes (polygenic) .**
b. **The polygenes affect the trait in the same way as an additive fashion.**
c. **Environment has a large effect on such phenotypes. Environmental variations will tend to smooth out the differences between phenotypic groups so providing continuous variations.**
d. **Crossing over during prophase I of meiosis between the alleles of non-sister chromatids of homologous chromosomes may also increase the recombination of alleles in the individual**
e. **The independent assortment and segregation of chromosomes during metaphase I of meiosis ensures that individuals possess a range of genotype from any polygenic complex. → This is because of different combination of maternal and paternal chromosomes from both parents;**
f. **Fertilisation is also a random process/ random fusion of gamete, with gametes carrying different combination of alleles fusing with each other non-discriminately.**

(c) Suggest why percentage of mortality is higher on both ends of the range of birth weights of new-born Siberian husky puppies. [2]

a. **High birth weight → If these muscles were heavier / high birth weight, they would cause the puppies to sink into the snow, and cause it to move slower or get stuck in the snow. → cannot escape from predators.;;**
b. **Low birth weight → result in premature death → underdeveloped organs → higher mortality.;;**

(d) Suggest why Siberian husky and coyote are classified as different species. [2]

a. **They do not have similar morphological / anatomical / physiological features;;**
b. **thus they are incapable of interbreeding;**
c. **incapable of producing viable fertile offspring;**

(e) Some scientist studied the anatomical structures of the golden jackal and hypothesized that the golden jackal is more closely related to the Siberian husky than the grey wolf. Fig. 3.3 shows a segment of homologous DNA sequences from the golden jackal, coyote, grey wolf and Siberian husky.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
golden jackal	A	G	C	T	G	T	C	G	A	T	T	C	C	A
coyote	A	G	C	T	A	T	G	G	A	A	T	C	G	A
grey wolf	T	G	C	T	A	T	G	G	A	T	T	C	C	T
Siberian husky	T	G	G	T	A	T	G	G	A	T	T	C	C	A

Fig. 3.3

Suggest if the hypothesis that the golden jackal is more closely related to Siberian husky than the grey wolf is true. [2]

a. The hypothesis is incorrect / not true;

b. Siberian Husky and grey wolf have 2 nucleotide differences (nucleotide 3 and 14) while golden jackal and Siberian Husky have 4 nucleotide differences(nucleotide 1, 3, 5, 7);;

c. This indicates that the dog has a greater degree of homology in DNA sequence with grey wolf than with golden jackal;

OR

d. indicating that the dog and grey wolf share a more recent common ancestor.

Max 2m

[Total: 12]

Section B

Answer **ONE** question in this section.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clear labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

4 **(a)** With reference to named examples, describe the range of roles performed by the proteins in living organisms. [13]

(b) Describe how Southern blotting can be used to analyse nucleic acids. [12]

5 **(a)** With reference to named examples, describe the range of roles performed by ATP in living organisms. [13]

(b) Compare and contrast between oxidative phosphorylation and photophosphorylation. [12]

4 (a) With reference to named examples, describe the range of roles performed by the proteins in living organisms. [13]

a) **enzymes with role in catalyzing chemical reactions;**
   i. **Adenylate cyclase catalyses the conversion of ATP to cAMP**
b) **description of at least two different types of reactions catalyzed by enzymes e.g. hydrolysis, polymerization, phosphorylation, oxidation / reduction, etc,**
   i. **GTPase, which hydrolyses its bound GTP on G protein to GDP → causes G protein to be inactive**
   ii. **activated protein kinase A activates a large number of phosphorylase kinase**
   iii. **Glycogen phosphorylase catalyses the conversion of a large number of glycogen to glucose-1-phosphate**
   iv. **A tyrosine kinase is an enzyme that catalyses the transfer of phosphate groups from ATP to tyrosine (amino acid) residues on a substrate protein, activating it.**
c) **name example of enzyme with reaction catalyzed;**
d) **hormone with role as chemical messenger**
   i. **insulin → binding to insulin receptors RTK ,**
   ii. **glucagon → binding to G-protein coupled receptors**
e) **Named example of hormone with specific role**
   i. **insulin and glucagon → regulate blood glucose levels**
f) **antibodies / immunoglobulins and role, e.g. bind to antigens / agglutinate pathogens etc;**
g) **transcription factors (activators and repressors) → to regulate transcription**
h) **haemoglobin → for oxygen transport;**
i) **carrier / channel proteins with role in passive transport;**
j) **carrier proteins with specific role**
k) **pumps with role in active transport**
l) **pump with role**
m) **further example of membrane transport protein with contrasting role**
n) **electron transport chain, components for chemiosmosis / ATP synthesis / redox reactions;**
o) **tubulin for flagella, cilia, spindle, cytoskeleton**
p) **collagen for structure / strength / bone / skin / connective tissue;**
q) **G protein for signal transduction at cell surface membrane**
r) **histones for packaging DNA;**
s) **AVP e.g. role and named / class of protein**
t) **glycoprotein and roles → cell surface receptors, cell-cell recognition / attachment in viral coats / capsids.**

QWC communicated clearly without ambiguity to include 6 different roles. (1m)
1m for each role , 1m for specific examples (6 different roles to be covered)

(b) Describe how Southern blotting can be used to analyse nucleic acids. [12]

Gel electrophoresis (4m) +2m

a) Genomic DNA is cut by restriction enzymes into DNA fragments

b) DNA fragments (obtained after digestion by restriction enzyme) are loaded in a well nearest to the negative electrode of the agarose gel.

c) A DNA ladder is also loaded in another well. (DNA ladder contains DNA fragments of known sizes for visual comparison with the sample fragments);;

d) DNA being negatively charged, due to its phosphate groups, will move towards the positive electrode /anode

e) once the electric current is switched on.

f) DNA ladder at the first and last lanes to allow determination of the sizes of the DNA fragments from the sample DNA

g) A loading dye (Bromophenol blue and glycerol) is added to the solution containing the DNA fragments → assist in loading of DNA and gives indication of how long gel electrophoresis have taken place and where the DNA molecules have migrated.;;

h) A buffer solution is used to maintain proper pH and ion concentration as well as to provide the necessary electrolytes for conducting electricity.

i) Allow gel electrophoresis to run and stop when the dye front reach 2/3 of the gel → to prevent any small DNA fragments/bands from running out of the gel.

j) Distance travelled by the DNA bands in a given time depends on the molecular mass/weight of the DNA as the fragments have to maneuver through the pores of the gel

k) Larger DNA fragments travelling slower and smaller fragments travelling faster → larger fragments, more resistance so slower rate of movement

(2m)

~~l) Ethidium Bromide (staining dye) and UV light/ fluorescent dyes are used to visualise the DNA bands. (may also suggest use of radioactivelly labelled probes and use of autoradiography to detect – Southern blotting) ;;~~

~~m) Gel electrophoresis enables the separation & visualisation of DNA fragments obtained from restriction digestion of alleles present in the sample;;~~

Southern blotting (2m)

n) where a sheet of either nitrocellulose paper or nylon paper is laid over the gel, and the separated DNA fragments are transferred to the sheet via capillary action;

o) DNA denaturation takes place which disrupts hydrogen bonds between complementary nucleotide bases;

p) causes the double helix of DNA to be separated into two molecules of single-stranded DNA;

q) using an alkaline solution.

Nucleic acid hybridisation; (3m)

r) Radioactively labeled single-stranded DNA/RNA probe;

s) anneals / hybridizes to ssDNA at the region complementary to the sequence of probe;

t) hydrogen bonds form via complementary base-pairing to the gene of interest

u) Finally the nitrocellulose membrane is subjected to autoradiography;

v) The DNA which the radioactively labeled probe binds to will show up as bands on the autoradiograph.;;

w) This yields a band pattern characteristic of gene of interest;

x) Nucleic acid hybridization allows us to determine the size of the band(s) containing the gene of interest.

y) Required for analysis of alleles of a particular genes → differentiate the alleles based on the DNA bands / gene of interest obtained

z) Also allow for the isolation of pure DNA fragments from a variety of bands

[1 – QWC as long as headings are seen]

5 **(a)** With reference to named examples, describe the range of roles performed by ATP in living organisms. [13]

a. ATP – adenosine triphosphate, an energy currency in the cell produced by phosphorylation of ADP + P_i ;;

b. by substrate-level phosphorylation, oxidative phosphorylation, and photophosphorylation ;;

c. when hydrolysed into ADP + P_i, releases a lot of energy to fuel many <u>anabolic reactions</u> within the cell ;;

d. Choose any 8 reactions:

e. needed in the endomembrane system i.e. to supply energy needed to power the migration of vesicles (transport / secretory) between organelles for protein / lipid trafficking ;;

f. maintaining the ionic gradient inside and outside the cell / across the plasma membrane, by providing the energy to pump 3 Na^+ out, and 2 K^+ in via the Na^+/K^+ ATPase / pump ;;

g. required for active transport processes i.e. to move substances against concentration gradient through carrier proteins / in bulk transport processes, endocytosis and exocytosis ;;

h. synthesis of large biomolecules (proteins, carbohydrates) through formation of bonds between monomers (peptide bonds, glycosidic bonds / amino acids, monosaccharides) requires energy ;;

i. phosphorylation of proteins, where a phosphate group from ATP is added to proteins → activating / inactivating proteins by triggering a 3D conformational change → e.g. kinases in signal transduction pathways ;;

j. phosphorylation of (RTK) receptors → phosphate groups then act as docking sites for the recognition and binding of relay proteins to trigger signal transduction ;;

k. can be used to phosphorylate GDP to GTP, allowing for GTP to be used as an energy currency to aid in ribosome function during translation (codon recognition / ribosomal translocation) ;;

l. polymerisation of microtubules during cell division requires ATP → MTs can then be attached to kinetochores of chromosomes to align them at the metaphase plate / elongation of the cell ;;

m. act as an allosteric inhibitor in respiration i.e. high ATP inhibits glycolysis (phosphofructokinase enzyme) / Krebs cycle ;;

n. energy from ATP is required in the formation of G3P in Calvin cycle, as well as in the regeneration of RuBP during light independent stage ;;

o. **ATP/AMP ratio acts as a biosensor in bacteria → low levels of ATP corresponding to high levels of AMP triggers increased transcriptional rates of the *lac* operon ;;**

QWC (2m) points communicated clearly without ambiguity and with relevant examples as to how ATP is useful in living organisms (plants, animals, prokaryotes)

(b) Compare and contrast between oxidative phosphorylation and photophosphorylation. [12]

Similarities

a) **Require ~~protein complexes and~~ (mobile) electron carriers that are embedded in membranes ;;**

b) **Requires the flow of electrons down its energy gradient from an electron donor to a final electron acceptor ;;**

c) **Require the pumping of H^+ ions across membranes to generate a proton motive force / region of high H^+ electrochemical gradient / proton gradient ;;**

d) **Require the coupling of the exergonic flow of H^+ down its electrochemical gradient to provide energy for phosphorylation of ADP to ATP via chemiosmosis ;;**

e) **The phosphorylation of ADP to ATP is mediated by ATP synthase complex ;;**

Differences

Features	Photophosphorylation	Oxidative Phosphorylation
f) Location ;;	occurs on thylakoid membranes of chloroplasts	occurs on inner membranes of mitochondria
g) Requirement of light energy ;;	light energy required for splitting of water (to donate electrons)	independent of light energy
h) Number of pathways ;;	2 pathways i.e. cyclic or non-cyclic pathways	1 pathway
i) Identity of electron donors ;;	Water (in NCP) AND PSI (in CP)	Reduced NAD and FAD
j) Identity of final electron acceptor ;;	NADP (in NCP) AND PSI (in CP)	Oxygen
k) Location of proton gradient ;;	Between stroma and thylakoid space	Between matrix and intermembrane space

l) Usage of ATP produced ;;	ATP produced is used within chloroplast (in light independent reaction)	ATP produced is used throughout the cell
m) Metabolic process ;;	Photophosphorylation occurs as part of an anabolic process i.e. production of sugars	Oxidative phosphorylation occurs as part of a catabolic process i.e. breakdown of sugars

QWC(a) ;; answers address both similarities and differences (at least 2 of each)

QWC(b) ;; answers are organised into two headers (similarities and differences), with point-to-point comparison being made

END OF PAPER